ELSEVIER

Precambrian Research 100 (2000) 335–357

Precambrian Research

www.elsevier.com/locate/precamres

Age and palaeomagnetism of the Mundine Well dyke swarm, Western Australia: implications for an Australia–Laurentia connection at 755 Ma

Michael T.D. Wingate [a,*], John W. Giddings [b]

[a] *Tectonics Special Research Centre, Department of Geology and Geophysics, The University of Western Australia, Nedlands, WA 6907, Australia*

[b] *Australian Geological Survey Organisation, Canberra, ACT 2601, Australia*

Abstract

Integrated U–Pb geochronology and palaeomagnetic studies of the Mundine Well dyke swarm (MDS) of the Pilbara Craton, Western Australia, are employed to test the hypothesis that Australia and Laurentia were joined as part of the Rodinia supercontinent. Ion microprobe U–Pb dating of zircon and baddeleyite indicates that the MDS was emplaced at 755 ± 3 Ma (95% confidence interval) and palaeomagnetic results confirm that the swarm is equivalent in age to dykes of the Northampton Inlier. A positive contact test between an MDS dyke and an older dolerite sill indicates that the MDS magnetisation dates from the time of dyke emplacement and cooling at 755 Ma. Combining new palaeomagnetic data for eight MDS dykes with previous results for six Northampton dykes yields a palaeopole for Australia at 135°E, 46°N ($A_{95}=4°$). The MDS palaeopole lies 30° away from the 780 to 740 Ma APW path segment for Laurentia, indicating that, if the Rodinia reconstruction is correct, breakup between Australia and Laurentia occurred prior to 755 Ma. It has been suggested elsewhere that the Sturtian glaciation in Australia preceded or accompanied breakup. If so, the Sturtian rocks should be older than 755 Ma and hence may not be coeval with glaciogenic rocks of the post-755 Ma Rapitan Group in Laurentia. Crown copyright © 2000 Published by Elsevier Science B.V. All rights reserved.

Keywords: Australia; Baddeleyite; Dyke; Neoproterozoic; Palaeomagnetism; Rodinia; Zircon

1. Introduction

The idea that Australia was joined to western Laurentia (Jefferson, 1978; Bell and Jefferson, 1987) is an integral part of most reconstructions of the Neoproterozoic supercontinent Rodinia (Dalziel, 1991; Hoffman, 1991; Moores, 1991; Ross et al., 1992; Brookfield, 1993; Powell et al., 1993; Borg and DePaolo, 1994; Gose et al., 1997), although alternative scenarios have been proposed (e.g. Li et al., 1996). If this aspect of the Rodinia reconstruction is correct, palaeomagnetic poles or apparent polar wander (APW) path segments of precisely the same age from each continent should coincide. Of the several palaeopoles for Neoproterozoic rocks of Australia, however, not one is dated precisely, and there are no reliable palaeomagnetic data for the interval ~1050 to ~700 Ma, during which Australia and Laurentia were postulated to have been joined as part of the

* Corresponding author. Fax: +61-8-9380-1090.
E-mail address: mwingate@tsrc.uwa.edu.au (M.T.D. Wingate)

0301-9268/00/$ - see front matter. Crown copyright © 2000 Published by Elsevier Science B.V. All rights reserved.
PII: S0301-9268(99)00080-7

Rodinia supercontinent (Powell et al., 1993, 1994). Veevers et al. (1997), based on tectonostratigraphic analysis, proposed that breakup between Australia and Laurentia did not occur until 560 Ma. Our new U–Pb and palaeomagnetic data for the 755 Ma Mundine Well dyke swarm (MDS) provide a precisely dated primary palaeopole for the Neoproterozoic of Australia and permit a rigorous test of the proposed Rodinia connection, at 755 Ma, between Australia and Laurentia.

2. The Mundine Well dyke swarm (MDS)

Archean and Proterozoic rocks of the Pilbara Craton and surrounding regions are intruded by an extensive swarm of dolerite dykes termed the MDS by Hickman and Lipple (1978). Most dykes are concentrated in the Gascoyne Complex and western parts of the Hamersley, Ashburton, and Bangemall Basins, where they trend mainly NNE (Fig. 1). The swarm swings eastward across granitoid-greenstone terranes of the Pilbara Craton, where it consists of four or five subparallel dykes, ~50 km apart, with individual strike lengths of over 200 km. Most dykes are less than 30 m wide and fine- to medium-grained, although some are wider than 300 m and coarse-grained. Both quartz- and olivine-normative types are present. The dykes represent the last known igneous event in the region, and are not metamorphosed. Several dykes are contaminated extensively by granitic xenoliths and some have metamorphosed and partially remelted their granitoid wall-rocks.

No geochronology has been undertaken previously on the MDS and its age is constrained poorly by ages of intruded and overlying rocks. The MDS postdates regional folding of extensive dolerite sills intruded into, and deformed with, shallow-marine sedimentary and minor volcanic rocks of the Bangemall Basin, deposited between ~1.64 and 1.05 Ga (Williams, 1990, and references cited therein; Nelson, 1995). MDS dykes are not known to intrude Devonian and younger strata of the Carnarvon Basin (Fig. 1) that unconformably overlie Palaeoproterozoic metamorphic and granitoid rocks of the western Gascoyne Complex (Condon, 1955, 1962; Hocking, 1990), but no contacts with the younger rocks have been observed.

3. Sample descriptions

Surface samples for U–Pb geochronology were obtained from four dykes of the MDS. Sample B6 (23°16.25′S, 115°46.23′E) was collected from a poorly-exposed NNE-trending olivine gabbro dyke intruded into the Irregully Fm. of the lower Bangemall Group. The sample is medium- to coarse-grained, with intergranular texture, and contains plagioclase, clino- and ortho-pyroxene, olivine, and minor magnetite, pyrite, interstitial quartz-feldspar granophyre, and baddeleyite. Alteration is minimal; secondary minerals include biotite, serpentine, amphibole, and chlorite.

Sample B7 (22°55.10′S, 115°57.50′E) is from the centre of a prominently exposed, 25 m wide, NE-trending dyke intruded into Palaeoproterozoic rocks of the Ashburton Basin and Gascoyne Complex (this outcrop area is also palaeomagnetic site I, as described below). The rock is a medium- to coarse-grained quartz dolerite, displays intergranular to subophitic textures, and contains clino- and ortho-pyroxene, plagioclase, interstitial quartz-feldspar granophyre, interstitial quartz, and minor magnetite, baddeleyite, and zircon. Plagioclase is moderately sericitised, augite altered partially to chlorite and biotite, and orthopyroxene largely replaced by talc, serpentine, and chlorite. B7 zircons range in length from 80 to 250 μm, and have length-to-width ratios up to 3:1. Most contain central channels filled with other minerals, suggesting rapid crystallisation. The zircons luminesce weakly, or not at all, using cathodoluminescence techniques, and no internal structures (other than concentric zoning in some crystals) are visible.

Samples B15 (21°21.32′S, 118°52.90′E) and B20 (20°53.03′S, 119°05.83′E) are from ENE-trending dykes intruded into Archean granite-greenstone terranes in the northern part of the Pilbara Craton. The two samples are similar petrographically, and consist of a fine-grained reddish-brown groundmass containing abundant corroded and embayed quartz and feldspar xenocrysts and medium- to coarse-grained granitic xenoliths in variable stages of assimilation. The groundmass is composed mainly of euhedral plagioclase, secondary chlorite, and abundant interstitial quartz-feldspar granophyre. The samples are weathered intensely, and

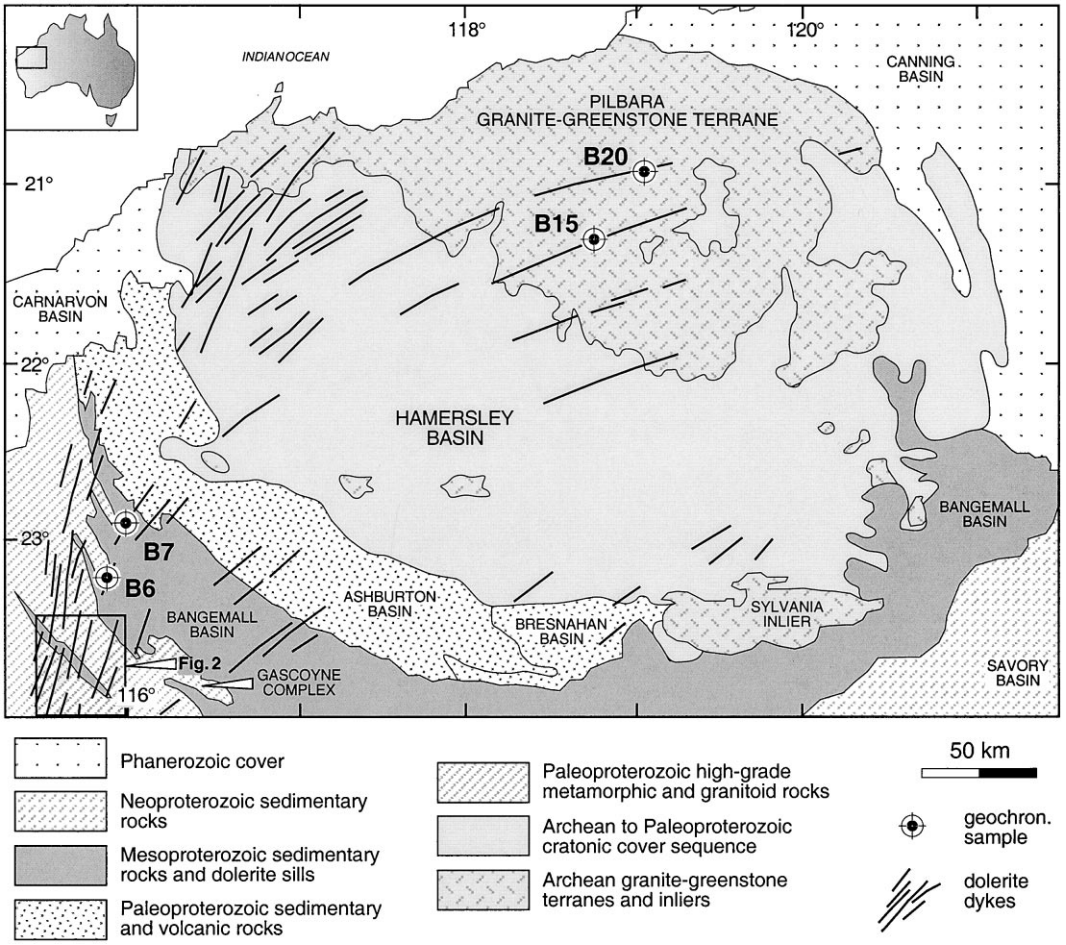


Fig. 1. Simplified geology of the Pilbara Craton and overlying and surrounding basins, showing the distribution of dykes of the MDS and sample locations therein. Except for ENE-trending dykes in the northern Pilbara Craton, many more dykes exist than are shown; lines indicate general dyke trends (compiled from maps of the 1:250 000 Geological Series published by the Geological Survey of Western Australia). The box in the SW corner indicates the area from which palaeomagnetic samples were obtained, shown in Fig. 2.

all minerals, except for large feldspar xenocrysts and quartz, are haematised strongly. Interiors of feldspars are altered to montmorillinite ± haematite. Other secondary minerals in the groundmass include amphibole, biotite, and epidote. A sample collected ~100 km to the WSW, from the same dyke as B15, consists of uncontaminated medium- to coarse-grained dolerite. More than 120 rounded, subhedral, and euhedral zircons were separated from each sample; the majority are strongly discoloured and cracked.

Samples for palaeomagnetic study were collected, using a portable diamond drill, from nine sites in eight MDS dykes (Fig. 2). Samples for baked-contact tests were drilled from host granite at sites A, C, and H, and from a dolerite sill intruded into, and deformed with, Bangemall Group sedimentary rocks at site E. At site I, samples were collected from the same outcrop that yielded geochronology sample B7. Coordinates of each site were determined using a GPS receiver. Cores were oriented using both a magnetic and a solar compass. Angular differences between the two measurements exceeded 5° for almost 50% of samples; in these cases sun readings were taken as being more accurate. All dyke samples are quartz

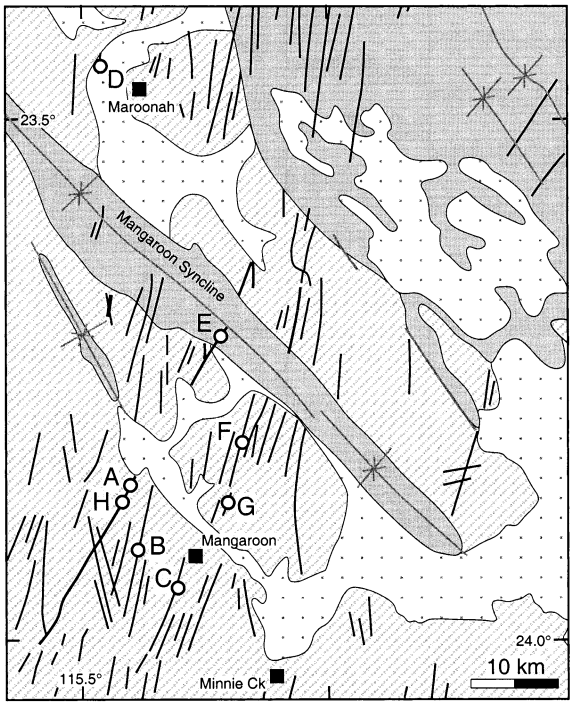


Fig. 2. Palaeomagnetic sample sites A to H in MDS dykes. See Fig. 1 for map legend and location (B7) of site I. Synclinal axes in folded Bangemall Group rocks are indicated by grey lines. Pastoral stations are shown by black squares. Compiled from Muhling and Brakel (1985) and 1:250 000 Geological Series maps.

dolerites, exhibit intergranular to subophitic textures, and contain augite ± pigeonite, plagioclase, interstitial quartz and quartz-feldspar granophyre, minor magnetite and apatite. Alteration is slight to moderate, and is variable between dykes and between cores from the same dyke; plagioclase is locally sericitised, particularly along fractures, and pyroxenes are partially altered to bluish-green amphibole and chlorite. Orthopyroxene in some samples is subordinate to clinopyroxene and is partially replaced by talc, serpentine, and chlorite.

4. U–Pb geochronology

Zircons from sample B7 were analysed during a single session on the SHRIMP II ion microprobe at the Research School of Earth Sciences, Australian National University (ANU), using standard operating procedures (Claoué-Long et al., 1995). U–Th–Pb ratios and absolute abundances were determined by comparison with the ANU AS3 zircon standard. Concentrations of U and Th in B7 zircons range as high as 6700 ppm and 18000 ppm respectively, and Th/U ratios vary between 1.3 and 7.5 (Table 1). The proportion of common ^{206}Pb to total ^{206}Pb, estimated using ^{204}Pb, is less than 0.6% for all analyses. The $^{207}Pb^*/^{206}Pb^*$ ratios (Pb^* denotes radiogenic Pb) measured in 23 B7 zircons agree within error and form a single, normally distributed population with a mean of 754.8 ± 1.2 Ma (1σ).

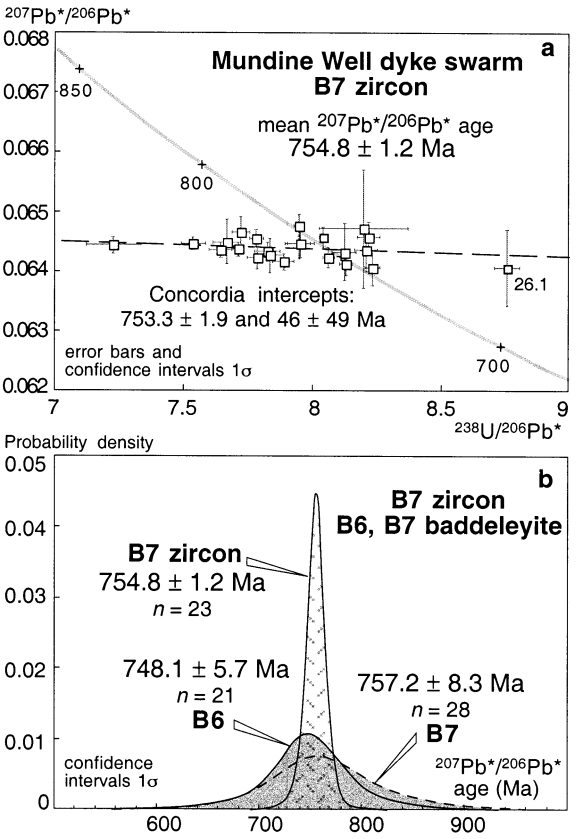


Fig. 3. (a) U–Pb evolution diagram showing SHRIMP analytical results for B7 zircons. Concordia intercepts were determined by regression of all data to yield a well-fitted (MSWD = 0.79) discordia. (b) Probability density plot of $^{207}Pb^*/^{206}Pb^*$ ages obtained for B6 and B7 baddeleyite, compared with those for B7 zircon.

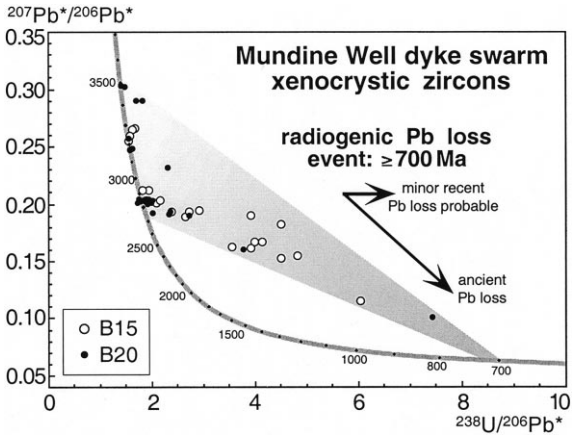


Fig. 4. U–Pb evolution diagram illustrating analytical data for xenocrystic zircons from samples B15 and B20.

Corresponding $^{206}Pb^*/^{238}U$ ratios, however, are dispersed beyond analytical precision [Fig. 3(a)], and yield ages ranging from ~740 to 835 Ma (excluding analysis 26.1 that, at 698 Ma, appears to have lost Pb). Two analyses are slightly normally discordant (and may have undergone minor Pb loss), but more than half are strongly reversely discordant. Good correlation is observed between apparent $^{206}Pb^*/^{238}U$ age and U concentration, and lower $^{206}Pb^*/^{238}U$ ages are in agreement with the mean $^{207}Pb^*/^{206}Pb^*$ age. These observations suggest either the presence of unsupported radiogenic Pb (Mattinson et al., 1996), or enhanced sputtering of Pb relative to U due to radiation-induced microstructural changes (McLaren et al., 1994); low birefringence observed in some B7 zircons could be evidence in favour of the latter mechanism. Regression of all data [Fig. 3(a)] indi-

Table 1
Ion microprobe analytical data for zircons from dyke sample B7[a]

Grain area	U (ppm)	Th (ppm)	Th/U	f_{206} (%)	$^{206}Pb^*/^{238}U$ ($\pm1\sigma$)		$^{207}Pb^*/^{206}Pb^*$ ($\pm1\sigma$)		Apparent age (Ma)	($\pm1\sigma$)
Z2704 B7 zircon										
1.1	972	3664	3.77	0.573	0.1232	0.0007	0.064 33	0.000 49	752.5	15.9
2.1	1100	6254	5.69	0.038	0.1216	0.0004	0.064 07	0.000 29	744.0	9.7
4.1	1869	4448	2.38	0.051	0.1231	0.0004	0.064 13	0.000 20	746.0	6.5
5.1	4416	5797	1.31	0.003	0.1384	0.0020	0.064 44	0.000 13	756.0	4.2
8.1	3777	15631	4.14	0.009	0.1241	0.0004	0.064 23	0.000 18	749.3	5.8
9.1	3151	4989	1.58	0.013	0.1269	0.0007	0.064 16	0.000 13	747.0	4.4
10.1	2422	6286	2.60	0.042	0.1279	0.0008	0.064 34	0.000 16	752.7	5.3
11.1	3181	6661	2.09	0.004	0.1244	0.0003	0.064 57	0.000 18	760.3	5.8
12.1	3192	9343	2.93	0.007	0.1286	0.0005	0.064 55	0.000 15	759.8	4.8
13.1	1433	10743	7.50	0.092	0.1219	0.0006	0.064 37	0.000 36	753.8	11.9
14.1	3853	9349	2.43	0.003	0.1305	0.0009	0.064 50	0.000 37	758.0	12.0
15.1	5023	18253	3.63	0.204	0.1259	0.0003	0.064 77	0.000 18	767.0	5.9
16.1	2706	7777	2.87	0.203	0.1296	0.0006	0.064 67	0.000 24	763.6	7.8
17.1	1334	5409	4.06	0.111	0.1221	0.0026	0.064 72	0.001 00	765.1	32.1
18.1	1440	3430	2.38	0.021	0.1218	0.0006	0.064 57	0.000 26	760.3	8.5
19.1	4220	10910	2.59	0.006	0.1309	0.0008	0.064 37	0.000 12	753.7	4.1
20.1	1536	5951	3.87	0.133	0.1259	0.0007	0.064 46	0.000 25	756.8	8.2
21.1	6694	15562	2.32	0.003	0.1298	0.0003	0.064 38	0.000 11	754.2	3.7
22.1	1450	6455	4.45	0.111	0.1259	0.0009	0.064 47	0.000 23	757.2	7.5
23.1	6651	13524	2.03	0.007	0.1328	0.0008	0.064 47	0.000 11	756.9	3.5
24.1	2882	8700	3.02	0.056	0.1285	0.0007	0.064 24	0.000 19	749.7	6.3
25.1	2795	8002	2.86	0.408	0.1277	0.0009	0.064 27	0.000 29	750.5	9.5
26.1	1191	4134	3.47	0.576	0.1144	0.0007	0.064 07	0.000 64	744.1	21.0

[a] Analyses are listed in the order in which they were conducted. f_{206} is the proportion of common ^{206}Pb in total measured ^{206}Pb. Apparent ages are calculated from $^{207}Pb^*/^{206}Pb^*$ ratios.

Table 2
Ion microprobe analytical data for baddeleyite from dyke samples B6 and B7[a]

Grain area	U (ppm)	Th (ppm)	Th/U	f_{206} (%)	$^{207}Pb^*/^{206}Pb^*$ ($\pm 1\sigma$)		Apparent age	
							(Ma)	($\pm 1\sigma$)
Z2196 Session 1 B6 baddeleyite								
1.1	259	9	0.03	0.230	0.064 79	0.002 17	767.6	69.0
2.1	434	32	0.07	0.230	0.063 66	0.001 68	730.2	55.0
3.1	325	3	0.01	0.092	0.064 37	0.001 24	753.8	40.2
4.1	1283	66	0.05	0.103	0.064 76	0.000 61	766.6	19.8
5.1	682	10	0.01	0.080	0.064 72	0.000 72	765.3	23.3
6.1	1357	86	0.06	0.140	0.063 74	0.000 62	733.0	20.6
7.1	1358	83	0.06	0.098	0.064 48	0.000 52	757.4	16.8
8.1	220	13	0.06	1.029	0.064 70	0.002 10	764.7	66.9
9.1	154	4	0.02	0.390	0.063 43	0.002 37	722.5	77.4
10.1	369	5	0.01	0.132	0.064 51	0.001 81	758.4	58.0
Z2196 Session 1 B7 baddeleyite								
*1.1	384	73	0.19	0.361	0.061 20	0.001 31	646.2	45.5
2.1	122	2	0.01	0.378	0.067 38	0.003 52	849.6	105
3.1	101	4	0.04	0.268	0.064 62	0.002 03	762.0	64.8
4.1	136	3	0.03	0.605	0.064 86	0.002 18	769.7	69.3
5.1	407	63	0.16	0.427	0.064 57	0.000 98	760.2	31.8
6.1	663	57	0.09	0.161	0.064 03	0.001 13	742.7	36.8
7.1	128	6	0.05	0.198	0.066 14	0.002 27	810.9	70.3
*8.1	285	54	0.19	0.453	0.059 46	0.001 55	584.1	55.6
9.1	175	4	0.02	0.513	0.064 42	0.002 82	755.5	89.8
Z2196 Session 2 B6 baddeleyite								
1.1	1044	57	0.05	0.410	0.063 64	0.004 17	729.6	133
2.1	1149	12	0.01	0.253	0.064 88	0.001 28	770.5	41.1
3.1	1205	15	0.01	0.370	0.064 34	0.001 14	752.7	37.0
4.1	305	7	0.02	0.637	0.064 94	0.002 15	772.2	68.2
5.1	5749	244	0.04	0.138	0.063 78	0.000 43	734.3	14.1
6.1	1494	101	0.07	0.308	0.063 45	0.000 97	723.5	32.1
7.1	1123	31	0.03	0.252	0.063 84	0.000 82	736.4	27.1
8.1	940	38	0.04	0.334	0.064 03	0.001 41	742.5	45.8
9.1	1687	58	0.03	0.277	0.064 90	0.000 57	771.0	18.4
10.1	3332	132	0.04	0.120	0.064 13	0.000 43	746.0	14.0
11.1	1037	16	0.02	0.124	0.063 62	0.000 66	729.1	21.9
Z2196 Session 2 B7 baddeleyite								
1.1	817	113	0.14	0.812	0.064 51	0.001 98	758.3	63.6
2.1	146	6	0.04	2.944	0.067 78	0.003 10	861.8	92.0
3.1	395	42	0.11	0.654	0.064 75	0.002 43	766.3	77.2
4.1	442	8	0.02	0.938	0.065 41	0.001 56	787.5	49.3
*5.1	586	71	0.12	4.046	0.052 03	0.001 81	286.6	77.5
6.1	357	28	0.08	0.544	0.065 33	0.001 58	785.0	49.9
*7.1	811	60	0.07	2.482	0.059 42	0.002 12	582.7	75.7
8.1	156	5	0.03	4.437	0.061 65	0.003 07	662.0	103
9.1	417	34	0.08	0.303	0.064 06	0.001 39	743.7	45.1
10.1	1314	123	0.09	0.866	0.063 65	0.000 77	730.2	25.4
11.1	300	18	0.06	1.482	0.064 79	0.002 31	767.5	73.5
*12.1	1443	341	0.24	1.581	0.061 90	0.000 63	670.5	21.8

Table 2 (*continued*)
Ion microprobe analytical data for baddeleyite from dyke samples B6 and B7[a]

Grain area	U (ppm)	Th (ppm)	Th/U	f_{206} (%)	$^{207}Pb^*/^{206}Pb^*$ ($\pm 1\sigma$)		Apparent age	
							(Ma)	($\pm 1\sigma$)
Z1897 Session 2 B7 baddeleyite								
1.1	594	41	0.07	1.091	0.064 15	0.001 08	746.5	35.0
2.1	480	56	0.12	0.346	0.064 38	0.001 32	754.0	42.8
3.1	1772	116	0.07	0.259	0.064 21	0.000 84	748.5	27.3
4.1	272	12	0.04	0.515	0.064 61	0.001 74	761.6	55.8
5.1	392	14	0.04	0.411	0.064 23	0.001 39	749.1	44.9
6.1	214	24	0.11	3.262	0.062 53	0.003 09	692.4	102
7.1	207	5	0.03	1.050	0.065 57	0.001 72	792.8	54.0
8.1	843	63	0.07	0.278	0.064 67	0.000 81	763.5	26.3
9.1	1080	72	0.07	0.323	0.064 67	0.000 98	763.6	31.5
10.1	515	34	0.07	0.588	0.064 49	0.000 99	757.6	32.1
11.1	454	36	0.08	0.990	0.065 25	0.001 19	782.5	37.8
12.1	145	8	0.06	0.740	0.063 96	0.001 75	740.2	57.0

[a] U and Th concentrations are approximate, but are proportional to true values. Analyses not employed in calculating mean $^{207}Pb^*/^{206}Pb^*$ ratios are marked with asterisks. Other notes as in Table 1.

cates that any isotopic disturbance was geologically recent, and, on this basis, the weighted mean $^{207}Pb^*/^{206}Pb^*$ age of 755 ± 3 Ma (95% confidence interval) is taken as the best estimate of the age of B7 zircons.

Baddeleyite crystals from samples B6 and B7 were analysed together during two analytical sessions, using procedures described by Wingate et al. (1998). Owing to crystal orientation effects that bias $^{206}Pb/^{238}U$ ratios measured in baddeleyite by ion microprobe (Wingate, 1997; Wingate and Compston, in press), only $^{207}Pb^*/^{206}Pb^*$ data are reported here (Table 2). Accuracy was monitored during each session by repeated analysis of a baddeleyite standard characterised previously by both SHRIMP and conventional isotope dilution analysis. The proportion of common ^{206}Pb, estimated using the 208-method (Hinthorne et al., 1979), is less than 1% for most analyses; for 10 of 54 analyses, common Pb ranges between 1 and 4.4%. After rejecting five of 21 results for B7 that lie more than $2\sigma_i$ below the mean value, all $^{207}Pb^*/^{206}Pb^*$ ratios for each sample agree within error, as do their weighted means. Combined analyses for both sessions yield $^{207}Pb^*/^{206}Pb^*$ ages of 748.1 ± 5.7 Ma for B6 and 757.2 ± 8.4 Ma for B7 (both 1σ). These results are not significantly different from each other, nor from the more precise age determined from B7 zircon [Fig. 3(b)].

Zircons from samples B15 and B20 were cast in the same mount and analysed during a single analytical session on SHRIMP II (Table 3). U–Th–Pb ratios and absolute abundances were determined relative to the ANU SL13 zircon standard. It is apparent that zircons in both samples were incorporated from rocks of Archean age, and that most have lost variable amounts of radiogenic Pb (Fig. 4). Those analyses close to Concordia define three groups in B20 (2.8 to 2.9, 3.15 to 3.25, and 3.5 Ga), and two groups in B15 (2.85 to 2.95 and 3.2 to 3.3 Ga). These ages are in close accord with those inferred for granitoid-gneiss suites in the north Pilbara Craton (e.g. Bickle et al., 1983, 1989; Blake and McNaughton, 1984). The majority of zircons in each sample appear to have crystallised at ~ 2.85 Ga, suggesting they were derived from late, post-tectonic granitoids of that age (Bickle et al., 1989).

Because the xenocrystic zircons are a mixed population derived from geological units of different ages, regression of the data to obtain Concordia intercepts is not justified. Treated in a semi-quantitative manner, however, the data suggest that the time of Pb loss was approximately

Table 3
Ion microprobe analytical data for xenocrystic zircons from dyke samples B15 and B20[a]

Grain area	U (ppm)	Th (ppm)	Th/U	f_{206} (%)	$^{206}Pb^*/^{238}U$ ($\pm 1\sigma$)		$^{207}Pb^*/^{206}Pb^*$ ($\pm 1\sigma$)		Apparent age	
									(Ma)	($\pm 1\sigma$)
Z2196 B15 zircon										
1.1	460	331	0.72	0.022	0.6476	0.0038	0.256 07	0.000 56	3222.7	3.4
1.2	1185	158	0.13	0.477	0.2558	0.0038	0.190 32	0.002 99	2744.9	25.6
2.1	1558	84	0.05	0.015	0.5332	0.0122	0.203 43	0.000 33	2853.9	2.6
2.2	351	179	0.51	0.059	0.4220	0.0034	0.194 23	0.000 72	2778.3	6.0
3.1	1030	311	0.30	0.427	0.3429	0.0096	0.194 67	0.000 55	2782.1	4.6
4.1	1001	146	0.15	0.021	0.6004	0.0047	0.267 03	0.002 34	3288.7	13.7
5.1	1459	79	0.05	0.008	0.4846	0.0056	0.201 42	0.003 01	2837.7	24.2
6.1	2867	985	0.34	0.108	0.2422	0.0060	0.167 36	0.001 24	2531.4	12.4
7.1	786	187	0.24	0.111	0.5199	0.0038	0.212 34	0.000 54	2923.5	4.1
8.1	6957	1373	0.20	0.075	0.1657	0.0051	0.115 89	0.000 98	1893.8	15.1
9.1	3235	208	0.06	0.113	0.2508	0.0088	0.167 40	0.003 36	2531.8	33.3
10.1	2269	833	0.37	0.176	0.2081	0.0062	0.155 05	0.005 96	2402.3	63.9
11.1	672	170	0.25	0.673	0.2224	0.0057	0.183 42	0.004 56	2684.0	40.5
12.1	1113	200	0.18	0.023	0.5570	0.0072	0.212 77	0.000 45	2926.7	3.4
13.1	224	157	0.70	0.029	0.6390	0.0065	0.259 64	0.001 24	3244.5	7.5
14.1	316	445	1.41	0.056	0.6198	0.0071	0.265 02	0.002 65	3276.8	15.6
15.1	1042	75	0.07	0.133	0.4647	0.0046	0.204 27	0.003 35	2860.6	26.4
16.1	3842	251	0.07	0.059	0.2229	0.0077	0.153 21	0.000 25	2382.1	2.8
17.1	4258	1138	0.27	0.089	0.2832	0.0073	0.163 65	0.003 12	2493.7	31.8
18.1	1492	242	0.16	0.126	0.3707	0.0074	0.194 00	0.002 07	2776.4	17.4
19.1	571	353	0.62	0.197	0.3790	0.0046	0.189 54	0.000 80	2738.2	6.9
20.1	3861	334	0.09	0.095	0.2566	0.0046	0.162 51	0.000 74	2481.9	7.7
Z2196 B20 zircon										
1.1	1386	127	0.09	0.140	0.6843	0.0178	0.303 17	0.009 12	3486.5	45.8
2.1	1144	230	0.20	0.025	0.5031	0.0093	0.201 91	0.002 50	2841.7	20.0
3.1	4015	5626	1.40	0.038	0.4998	0.0167	0.193 05	0.002 52	2768.3	21.2
4.1	2601	820	0.32	0.004	0.5544	0.0042	0.204 03	0.000 36	2858.7	2.9
5.1	5533	1127	0.20	0.036	0.2655	0.0086	0.160 82	0.003 41	2464.3	35.4
6.1	1284	185	0.14	0.036	0.5902	0.1045	0.290 73	0.019 30	3421.5	99.5
7.1	4349	144	0.03	0.004	0.4256	0.0070	0.194 32	0.001 16	2779.1	9.7
8.1	3645	2061	0.57	0.029	0.4315	0.0092	0.192 00	0.002 03	2759.4	17.2
9.1	1252	390	0.31	0.207	0.3685	0.0056	0.190 93	0.001 05	2750.2	9.0
10.1	305	123	0.40	0.094	0.5006	0.0069	0.203 74	0.002 89	2856.4	22.9
11.1	535	243	0.45	0.044	0.7180	0.0071	0.303 44	0.000 92	3487.9	4.7
12.1	1177	76	0.07	1.482	0.5213	0.0057	0.205 10	0.004 18	2867.2	32.8
14.1	–	–	0.31	0.004	0.5702	0.0077	0.203 26	0.002 20	2852.5	17.5
15.1	3109	945	0.30	0.001	0.5808	0.0101	0.205 01	0.000 42	2866.5	3.3
16.1	1682	469	0.28	0.058	0.5403	0.0100	0.205 37	0.001 00	2869.4	7.9
17.1	3183	1509	0.47	0.011	0.5730	0.0137	0.203 23	0.001 20	2852.3	9.6
18.1	2121	138	0.07	0.024	0.4369	0.0038	0.232 49	0.002 51	3069.2	17.1
19.1	2208	669	0.30	0.014	0.5877	0.0058	0.202 15	0.003 14	2843.6	25.1
20.1	2390	924	0.39	0.014	0.5424	0.0047	0.203 90	0.001 99	2857.7	15.8
21.1	959	947	0.99	0.149	0.5269	0.0051	0.200 05	0.001 15	2826.6	9.4
23.1	525	33	0.06	0.052	0.6397	0.0041	0.248 36	0.000 78	3174.4	4.9
24.1	776	59	0.08	0.031	0.6190	0.0055	0.248 44	0.001 23	3174.9	7.8
25.1	6887	459	0.07	0.256	0.1348	0.0037	0.101 63	0.000 38	1654.0	6.9
25.1	914	128	0.14	0.030	0.6530	0.0106	0.257 65	0.002 05	3232.4	12.5
1.2	323	145	0.45	0.368	0.5513	0.0063	0.290 30	0.005 18	3419.2	27.5
22.1	579	112	0.19	0.082	0.5072	0.0067	0.202 37	0.000 55	2845.4	4.4

[a] Notes as in Table 1.

700 Ma. The event responsible for the majority of Pb loss in these zircons is inferred to be their inclusion in originally basaltic magma of the B15 and B20 dykes, implying that the age of the dykes is approximately 700 Ma. Because some zircons are likely to have also undergone geologically recent Pb loss (which would shift data to the right, parallel to the horizontal axis in Fig. 4), 700 Ma should be regarded as a minimum age for dyke intrusion, and the true age is inferred to be slightly greater.

Agreement between baddeleyite and zircon results in samples B6 and B7 indicates that the zircons formed during dyke crystallisation, hence the $^{207}Pb^*/^{206}Pb^*$ age of 755 ± 3 Ma obtained for B7 zircons can be interpreted as an unambiguous age of emplacement for the B6 and B7 dykes. Patterns of Pb loss observed in xenocrystic zircons from B15 and B20 are consistent with the ENE-trending dykes of the north Pilbara region having been emplaced during the same event as the B6 and B7 dykes in the western part of the area. The best estimate of the age of emplacement of the MDS is 755 ± 3 Ma (95% confidence interval).

This result is not significantly different from a K–Ar age of 748 ± 8 Ma (1σ) reported by Embleton and Schmidt (1985) for an NE-trending dyke intruded into Mesoproterozoic metamorphic rocks of the Northampton Inlier, located ~450 km to the south (Fig. 5). Two other dykes from the same swarm yielded ages of ~550 Ma, which Embleton and Schmidt (1985) regarded as a minimum age of intrusion, suggesting that argon loss may have occurred during tectonism along an adjacent fault. Petrographically, the Northampton dykes are similar to the MDS, and both quartz and olivine

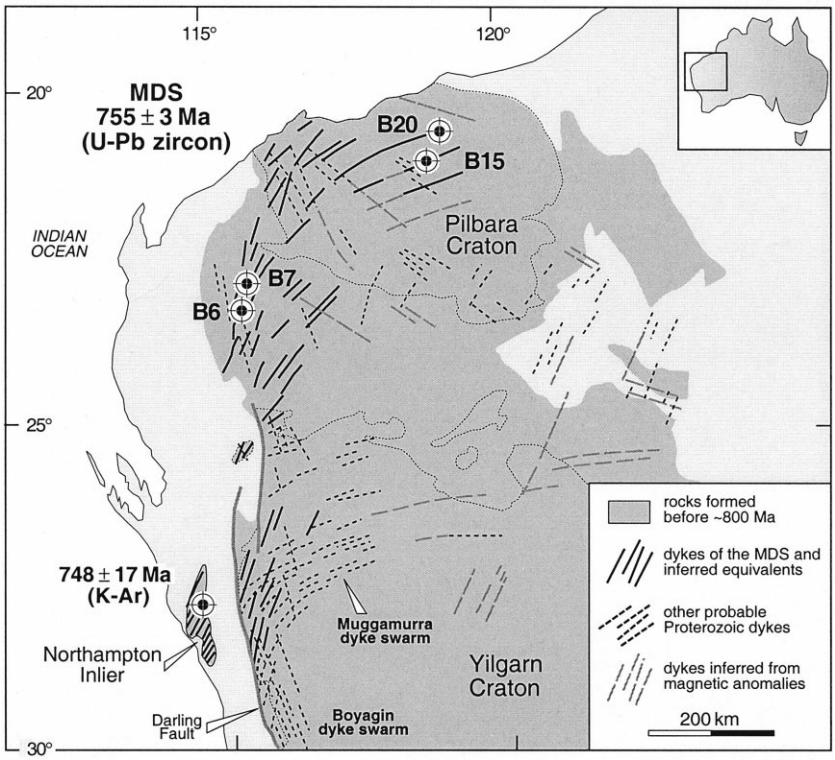


Fig. 5. Simplified map showing the distribution of the MDS and other dyke swarms of Meso- to Neo-proterozoic age in northwest Western Australia [compiled from Parker et al. (1987), Boyd and Tucker (1990), and 1:250 000 Geological Series maps]. Thin dotted lines indicate approximate limits of Archean cratons. The K–Ar age ($\pm 2\sigma$) for dykes of the Northampton Inlier is from Embleton and Schmidt (1985). NW-trending dykes in the Pilbara Craton belong to the Round Hummock swarm (Hickman and Lipple, 1978), discussed in the text.

dolerites are present (Embleton and Schmidt, 1985). Based on similarity in age, mineralogy, and orientation, it is concluded that dykes of the Northampton Inlier (and probably the adjacent Yilgarn Craton) are the southern extension of the MDS. This correlation is confirmed by the palaeomagnetic data described below.

5. Palaeomagnetism

Measurements were conducted on 72 cores (110 specimens) from eight dykes, 35 cores (40 specimens) from host granite, and 25 cores (25 specimens) from a dolerite sill intruded by an MDS dyke at site E. ScT and 2G-Enterprises cryogenic magnetometers and a Molspin spinner magnetometer were used for remanence measurements. Remanence composition was determined by detailed stepwise alternating field (AF) demagnetisation (>14 steps, up to 160 mT) of one or two specimens from each core, using the automated static method on the 2G-Enterprises magnetometer described by Giddings et al. (1997). Very strong specimens were measured initially using the tumbling method and a Schonstedt GSD-5 AF demagnetiser. One specimen from each of four dykes and two specimens from the host granite at site A were subjected to detailed stepwise thermal demagnetisation (>20 steps, 120 to >580°C), using a Schonstedt TSD-1 thermal demagnetiser. Magnetisation vectors were isolated using Principal Component Analysis (Kirschvink, 1980). Results for specimens from the same sample were averaged and sample means given unit weight in statistical calculations for each dyke. Results from two sites (A and H) in the same dyke are combined as a single site in Table 4.

Natural remanent magnetisations (NRMs) varied widely in both direction and intensity. Very high initial intensities (30 to 255 A m^{-1}) of some dyke and sill specimens are indicative of lightning-

Table 4
Palaeomagnetic results for the MDS and Northampton dykes[a]

Site	Location			Direction				Palaeopole			
	Lat. (°S)	Long. (°E)	N (n)	D (deg)	I (deg)	k	α_{95} (deg)	Lat. (°N)	Long. (°E)	K	A_{95} (deg)
MDS dykes											
EDMA+H	23.85	115.55	19 (7)	015.7	41.1	75	3.9	40.2	134.5	–	–
EDMB	23.91	115.56	11	017.2	35.1	51	6.6	43.6	138.3	–	–
EDMC	23.96	115.61	10 (1)	006.2	38.6	64	6.1	43.9	123.6	–	–
EDMD	23.46	115.53	5	014.4	13.8	66	9.5	56.5	142.1	–	–
EDME	23.71	115.63	10 (3)	010.6	34.6	84	5.3	46.0	130.1	–	–
EDMF	23.81	115.67	3	015.8	48.3	759	3.7	35.6	131.9	–	–
EDMG	23.88	115.64	6	015.8	33.8	205	4.3	45.8	136.6	–	–
EDMI(B7)	22.92	115.96	7	018.5	44.9	46	8.4	38.3	137.2	–	–
Mean of MDS dykes			8 dykes	014.2	36.4	49	7.5	43.8	134.1	103	5.1
Northampton Inlier dykes (Embleton and Schmidt, 1985)											
NDD01	27.89	114.58	8	019.9	26.6	–	10.5	43.8	141.8	–	–
NDD02	27.88	114.57	8	013.4	23.4	–	4.8	47.8	134.3	–	–
NDD03	27.88	114.56	7	005.1	16.0	–	4.4	53.5	123.1	–	–
NDD06	27.83	114.71	11	014.9	34.4	–	4.6	41.1	133.5	–	–
NDD08	27.82	114.70	3	030.3	14.6	–	23.7	44.1	158.9	–	–
NDD09	27.83	114.67	8	009.8	22.3	–	9.3	49.4	129.6	–	–
Overall mean (both localities)			14 dykes	–	–	–	–	45.3	135.4	88	4.1

[a] $N(n)$, total number of samples given unit weight in calculation of dyke means (number of samples included from baked contact); D, declination (E of N); I, inclination (positive downwards); $k(K)$, Fisher's precision parameter; $\alpha_{95}(A_{95})$, semi-angle of cone (radius of circle) of 95% confidence about mean direction (pole).

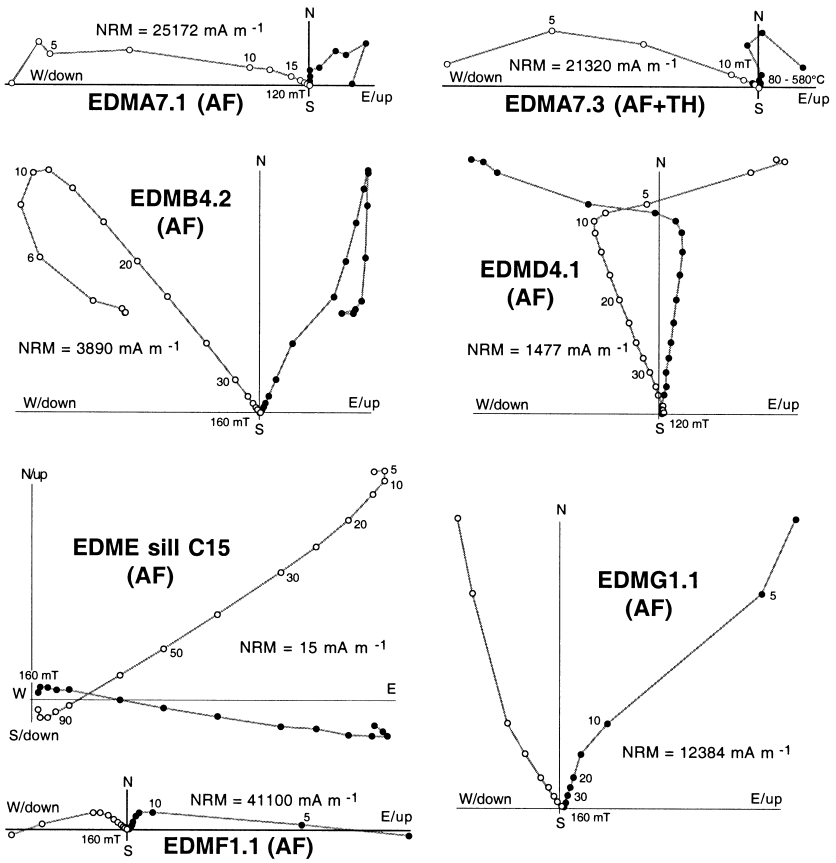


Fig. 6. Representative examples of demagnetisation behaviour for MDS dyke specimens, and for one specimen from the hybrid zone in the dolerite sill at site E (see text for explanation). Orthogonal projections show trajectories of the remanence vector endpoint during progressive demagnetisation (open/filled symbols represent projections on the vertical/horizontal plane). AF, alternating field; TH, thermal demagnetisation.

induced isothermal remanent magnetisation (IRM, Graham, 1961). For the majority of dyke specimens, a low coercivity (L) component was removed below 10 to 20 mT. This component is consistent in direction between specimens from the same core sample, but, in most cases, varies randomly between specimens from different samples, and is not similar to the present Earth's field (PEF) nor the geocentric axial dipole (GAD) field. The high intensities and dispersion suggest that the L component is primarily an IRM due to variable effects of lightning.

After removal of low-coercivity overprints, magnetisations in dyke specimens showed little change in direction at higher AFs, and have linear trajectories to the origin on orthogonal vector plots (Fig. 6). Most specimens yielded a consistently-directed, high-coercivity (H) component oriented slightly east of north with shallow to moderate downwards inclination. Following AF pretreatment to 15 or 20 mT (to reduce lightning-induced IRM, Schmidt, 1993), thermal demagnetisation yielded similar directions, with unblocking temperatures in the range 570 to 580°C, confirming the remanence to be single component and magnetite to be the dominant magnetic carrier. This is supported by the variation of low-field susceptibility with temperature (measured in specimens from four dykes using an Agico KLY3 Kappabridge), that shows well-defined Hopkinson peaks close to

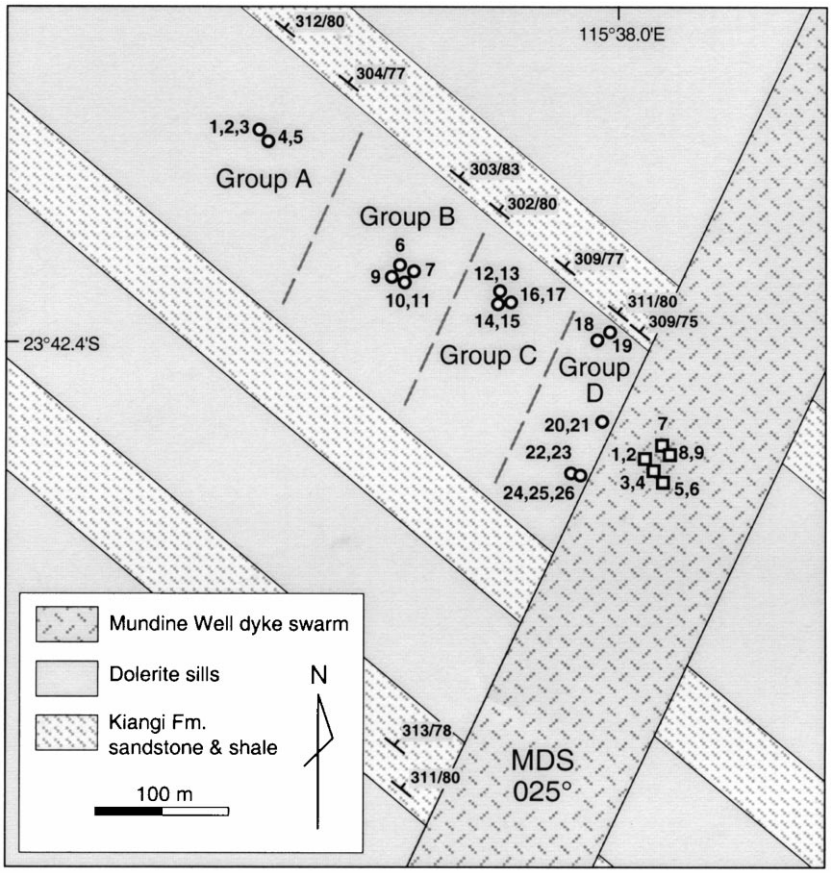


Fig. 7. Sample locations for the contact test between the MDS dyke and the host sill intruded into sediments of the Bangemall Group at site E, showing groups A to D discussed in the text. Bedding measurements consist of right-hand strike/dip.

580°C, indicative of the presence of fine-grained (single-domain, SD) magnetite (Schmidt, 1993). The overall mean direction for the H component (D, I) is 014°, +36°, $\alpha_{95}=8°$, with an associated palaeopole at 134°E, 44°N, $A_{95}=5°$ (Table 4).

Wingate (1997) reported that baked granite specimens at sites A and C are overprinted completely in the direction of dyke remanence, whereas granite specimens collected far from each dyke show consistently steep directions, implying a positive contact test. Subsequent measurement of samples from additional sites in the granite, however, has shown that the unbaked rock is generally unstable magnetically, and failed to isolate any coherent directions. Results from the granite, therefore, are not reported here, although overprint magnetisations measured in samples from baked contacts are included in calculation of dyke mean directions (details in Table 4).

A contact test at site E indicates that the H component of the dykes is of primary origin. At this site, a 125 m wide dyke crosscuts an outlier, within the Gascoyne Complex, of deformed Bangemall Group sediments and dolerite sills (Fig. 7). The sills alternate with sandstone and shale (Kiangi Fm.), and the strata are tilted 79° to the northeast (038°). Sampling was limited by the availability of in situ outcrop, and sill samples were collected in groups at four distances from the dyke contact. Samples in group A, collected furthest (350 m) from the dyke, yielded a single, well-defined, easterly-upwards direction at AF fields

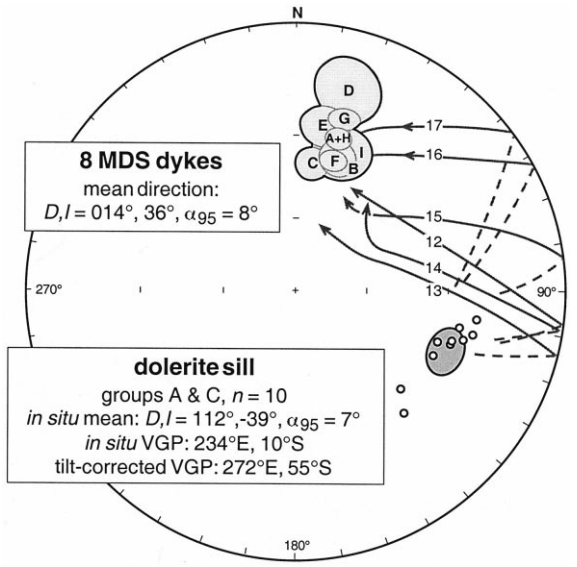


Fig. 8. Contact test results at site E. Mean palaeomagnetic directions for eight MDS dykes are shown by α_{95} confidence circles. Individual sample directions (small circles in upper hemisphere) and α_{95} confidence circle are shown for the dolerite sill. Trajectories (75 to 160 mT) towards the dyke direction are shown for group C sill samples; solid/dashed lines are in lower/upper hemisphere, numbers refer to samples in Fig. 6. Lower hemisphere equal area projection; primitive = present horizontal.

between 25 and 160 mT. No coherent directions were obtained from five samples in group B, 200 m from the dyke; they exhibited high intensities and became unstable between 10 and 50 mT. It is possible that some, if not all, group B samples have been remagnetised by lightning strikes. Six specimens in group C, ~120 m from the contact, exhibited two-component magnetisations. Similar easterly-upwards directions to those in group A were observed at AF fields up to 75 mT, after which the magnetisations rotated towards the dyke remanence direction, reaching it by 160 mT (Fig. 8). The magnetisations did not decay fully to the origin, even after treatment at 160 mT, suggesting the presence of fine-grained haematite. The trajectories of the magnetisations into the dyke remanence direction at higher AF fields imply that precipitation of haematite was associated with dyke intrusion. In the baked contact zone, within 30 m of the dyke (group D in Fig. 7), three of nine sill samples are overprinted completely in the dyke remanence direction. The remaining six samples possessed very high NRM intensities (30 to 117 A m^{-1}) and yielded scattered directions. These samples appear to have been almost completely overprinted by lightning-induced IRM, although three yielded incompletely-resolved directions similar to that of the dyke.

A positive baked contact test is thus constituted by: (1) agreement in direction between magnetisations in the dyke and the baked dolerite sill near its contact with the dyke; (2) a stable coherently-directed remanence in the unbaked dolerite sill, which is different in direction from that of the dyke (and in agreement, after tectonic correction, with the direction expected for an intrusion slightly older than the dyke, as described below); (3) the presence, at an intermediate distance from the contact, of a hybrid zone (Schwarz, 1977; Schwarz and Buchan, 1989) of partial remagnetisation of the sill, possessing both magnetisations. These observations indicate that magnetisation of the unbaked dolerite sill predates that of the dyke, and was overprinted in the contact zone during dyke emplacement. Preservation of both magnetisations indicates that the dyke magnetisation is primary, and dates from the time of dyke intrusion. A virtual geomagnetic pole (VGP) for the sill, after correction for the tilt of bedding, lies on the APW path for Australia between the MDS pole and those for the 1050 to 1090 Ma Kulgera and Stuart dykes (Fig. 9). This sill is one of several within the Bangemall Basin that are currently the subject of geochronological and palaeomagnetic investigation by the first author.

Additional evidence in favour of a primary origin for the MDS magnetisation lies in the presence of SD magnetite. Grains of this type are highly stable, and to reset their magnetisation in the geological environment would require temperatures approaching their Curie temperature of ~580°C (Pullaiah et al., 1975; Walton, 1980). The dykes, however, are not metamorphosed, and their emplacement is the youngest known igneous event in the region, hence it is unlikely that the dykes experienced post-emplacement temperatures sufficient to reset their original magnetisation completely. Based on this evidence, and the positive

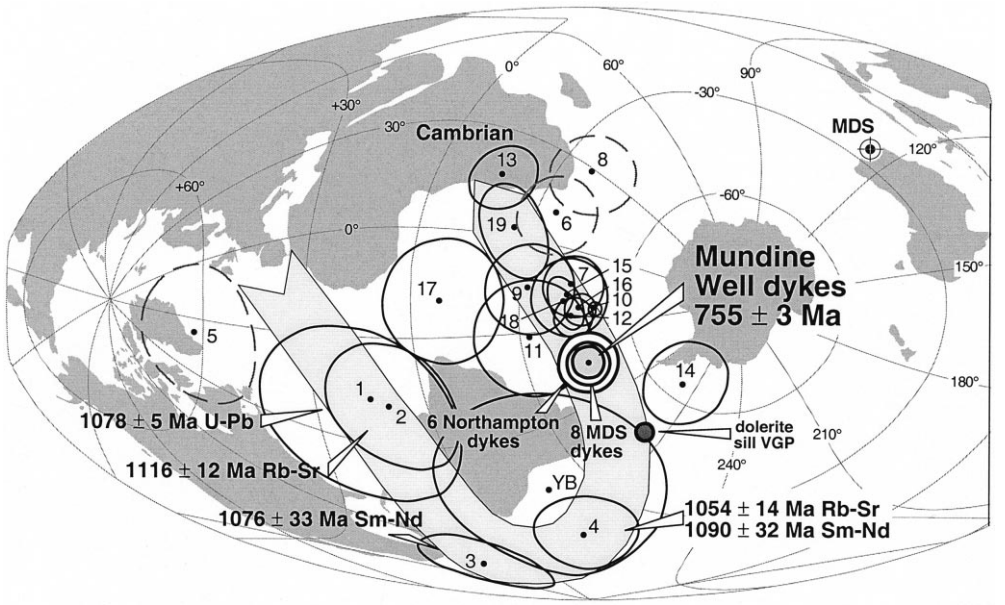


Fig. 9. Latest Mesoproterozoic to Early Cambrian south palaeopoles for Australia, with A_{95} confidence circles. Pole numbers refer to entries in Table 6. Results for the MDS (this study) and Northampton dykes (Embleton and Schmidt, 1985) are shown in bold; the mean and A_{95} confidence circle of the combined data are indicated by the shaded circle. Also shown is the bedding-corrected VGP for the unbaked dolerite sill at site E. Confidence circles of possible overprint poles are dashed. Aitoff projection centred on 75°E, 90°S, 40° clockwise azimuthal rotation.

contact test, we conclude that the H component is a primary thermoremanent magnetisation dating from the time of emplacement and cooling of the MDS at 755 ± 3 Ma.

The MDS pole is less than 4° from that determined for six dykes of the Northampton Inlier by Embleton and Schmidt (1985) (Table 4), confirming that the dykes of the Northampton Inlier are the southern extension of the MDS. Data from the two areas can be combined to yield a palaeopole, based on 14 dykes, at 135°E, 45°N, $A_{95}=4°$ (Fig. 9). Consistency between results for the MDS and those from the Northampton dykes implies that no significant relative movement between the two localities has occurred since 755 Ma (except in the improbable event that the palaeopole and Euler pole were coincident), and it is unlikely that two localities 450 km apart (Fig. 5) were tilted as a single block. In addition, dykes of the MDS, including those with NNE and ENE trends, transect relatively undeformed Archean to Palaeoproterozoic rocks of the Hamersley Basin, and, over most of the Pilbara region, cut across older (ca. 1600 Ma?) NW-trending dykes of the Round Hummock swarm (shown schematically in Fig. 5) without disturbing their linear NW trend (Hickman and Lipple, 1978; Tyler, 1990). Finally, all MDS dykes observed during this study dip within 5° of vertical and appear to be undeformed. From these observations, it appears likely that the MDS is in its original orientation, and that the eastward change in trend from NNE to ENE is a primary feature rather than the result of younger deformation. No tectonic correction has therefore been applied to the data.

The new MDS palaeopole, incorporating data from the Northampton dykes, is the first palaeopole for the Australian Neoproterozoic for which the age of magnetisation is known accurately and precisely. Although the MDS pole, at 755 Ma, is less than 15° away from younger Neoproterozoic to Cambrian poles (Fig. 9), it is up to ~200 million years older, suggesting either that the late Neoproterozoic was a period of minimal APW relative to Australia, or that the APW path is more complex than that determined from present data.

6. Discussion

6.1. Tectonic significance of the MDS

The MDS strikes for >900 km along the western and northern margins of the Western Australian shield (Fig. 5), and thus is likely to have been emplaced during a major tectonic event. Deep burial of Proterozoic rocks by those of Phanerozoic age along the western Australian margin has hindered efforts to determine the exact nature of this event. Reset mica Rb–Sr ages between ~700 and 800 Ma obtained for Palaeoproterozoic rocks of the Gascoyne Complex, interpreted in terms of either cooling during regional uplift or a brief thermal event at

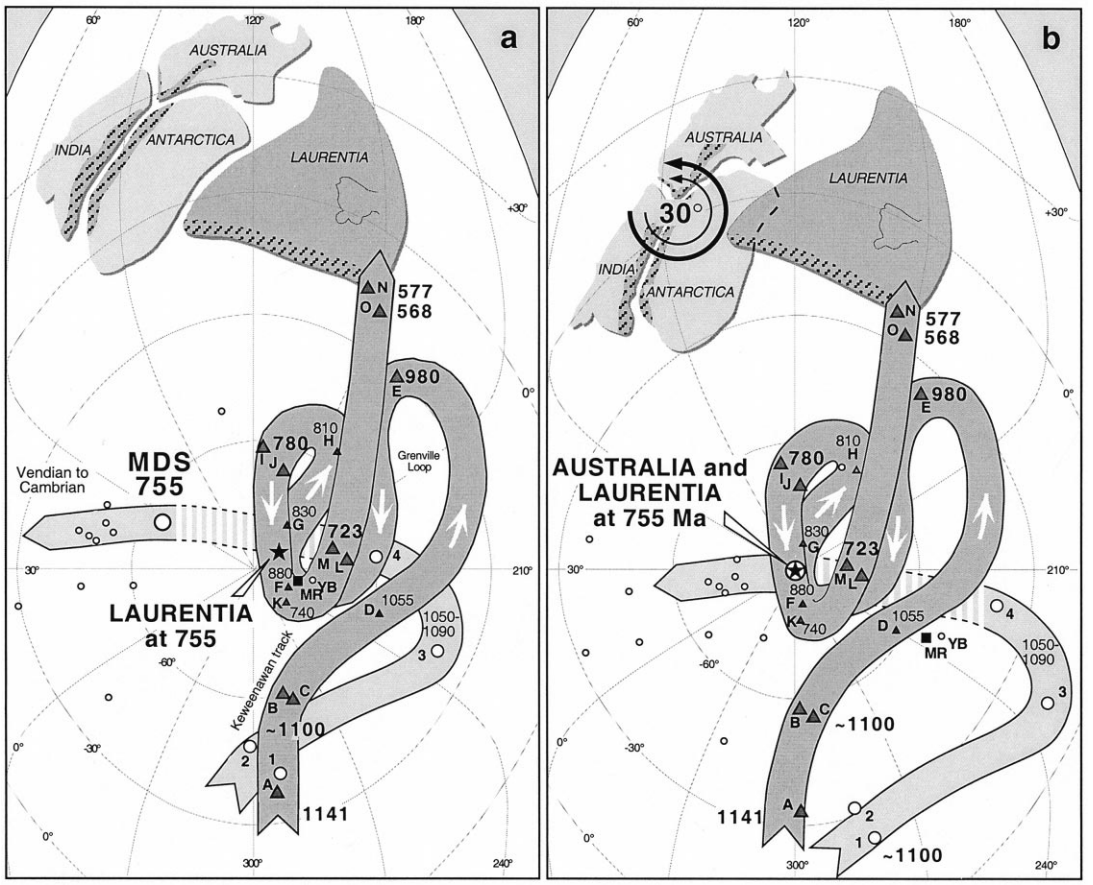


Fig. 10. Time calibration (ages in Ma) of latest Mesoproterozoic to Early Cambrian APW paths for Australia (circles) and Laurentia (triangles), with continents rotated into the Rodinia configuration of Powell et al. (1993). The APW path for Laurentia is based on many more data than are shown (e.g. Irving and McGlynn, 1981; Park and Jefferson, 1991; Hyodo and Dunlop, 1993; Park, 1994). An interpolated pole position for Laurentia at 755 Ma is shown by a star. The pole labeled MR is for the Malani Rhyolite, discussed in the text. Stippled areas of the continents indicate the approximate extent of Grenville-age orogenic belts [after Hoffman (1991) and Gose et al. (1997)]. Clockwise rotations to India (in present-day position): Australia, 64.75° about 15.35°E, 14.79°S; Antarctica, 92.45° about 17.14°E, 4.22°S; Laurentia, 166° about 339°E, 56°S (Powell et al., 1988, 1993). Aitoff projection centred on South Pole. (b) APW paths for East Gondwanaland and Laurentia rotated so that the MDS pole and the interpolated 755 Ma pole for Laurentia are superimposed at the south pole of the projection. The resulting configuration of the continents is shown above. East Gondwanaland is located further south and rotated ~30° relative to Laurentia from its position in (a); no exact location for the Euler pole is implied. Note that longitude is arbitrary, and that the continents and associated APW paths can be rotated around the projection pole along lines of palaeolatitude without violating palaeomagnetic constraints.

Table 5
Selected palaeomagnetic results from Proterozoic rocks of Laurentia[a]

Pole	Rock unit	Rock age (Ma)	Lat. (°N)	Long. (°E)	dp (deg)	dm (deg)	*N*	Comment	Reference
O*	Catoctin Volcanics	564±9 U–Pb 572±5 U–Pb	43	308	9	9	8	baked contact & fold tests	Meert et al. (1994), Aleinikoff et al. (1995)
N*	Callander Complex	577±1 U–Pb 575±5 K–Ar	46.3	301.4	5.9	6.1	29	reversal; contact test	Symons and Chiasson (1991) and references cited therein
M*	Franklin dykes	723+4/−2 U–Pb baddeleyite	−5	343	5	5	78	≥1 reversal	Heaman et al. (1992); mean pole from Park (1994)
L*	Natkusiak Fm.	723+4/−2 U–Pb baddeleyite	−6	339	4	8	14	reversal stratigraphy	Palmer et al. (1983), Heaman et al. (1992)
K	overprints in G & Thundercloud Fm.	~740±40	−22.5	341.3	8.8	8.8	22	hematite overprint absent in younger rocks	Park and Aitken (1986a,b); mean pole and age assignment from Powell et al. (1993), Park and Jefferson (1991)
J*	Tsezotene Sills & Dykes	779±2 U–Pb baddeleyite	−1.6	317.8	5	5	17	reversal; magnetite +deuteric hematite	Park (1981b,1989), Park et al. (1995) and references cited therein; Heaman et al. (1991)
I*	Wyoming Prov. dykes	782±8, 785±8 U–Pb baddeleyite	−13	311	3	6	9	crude contact test; regional consistency	Harlan et al. (1997)
H	Reynolds Pt Fm. & Little Dal Gp.	~810±40	11.0	324	–	–	74	–	Park (1981a), Palmer et al. (1983); mean pole and age assignment from Powell et al. (1993)
G	Tsezotene Fm. & Katherine Gp.	~830±50	−10.5	328.0	–	–	36	intruded by J	Park and Aitken (1986a,b); mean pole and age assignment from Powell et al. (1993)
F	Thanet Gabbro	~880	−20.3	339.3	11.7	11.7	18	CRM overprint	Buchan (1978), Park and Aitken (1986b)
E*	Haliburton Intrusions	980±10 Ar–Ar	36.0	322.5	6.3	6.3	42	–	Buchan and Dunlop (1976), Hyodo and Dunlop (1993)
D	Freda Sandstone & Nonesuch Shale	1055± ~25	−5.8	358.0	–	–	85	mag age pre-folding, but not established precisely	Henry et al. (1977); mean pole and age assignment from Powell et al. (1993)
C*	Logan Dykes	1109±4−2 U–Pb	−34.7	000.6	7.4	11.9	47	contact test (palaeointensity)	mean pole from Halls and Pesonen (1982); Davis and Sutcliffe (1985)
B*	Upper N. Shore Volcanics	1110±10 U–Pb	−32.4	003.8	3.8	6.0	98	fold & conglomerate tests	Pesonen (1979); mean pole from Halls and Pesonen (1982)
A*	Abitibi Dyke swarm	1141±2 U–Pb baddeleyite	−42.8	028.5	12.5	16.3	5	contact, secular variation tests	Krogh et al. (1987), Ernst and Buchan (1993)

[a] South Pole positions in present-day North American coordinates. Asterisks indicate key poles. Pole numbers identify data in Fig. 9. Lat. and Long., latitude (positive N) and longitude (E of N) of pole; dp and dm, semi-minor and semi-major axes of the oval of 95% confidence about the mean pole; *N*, number of samples averaged; CRM, chemical remanent magnetisation.

this time (Libby et al., 1986, and references cited therein), probably record the effects of uplift, extension, and heat advection that accompanied intrusion of the MDS. The observation that the dykes are subparallel to, and increase in density towards, the current continental margin suggests that their emplacement may be related to separation of an unknown continental fragment from Western Australia, possibly during plate reorganisation associated with the breakup of Rodinia.

Table 6
Palaeomagnetic results from Mesoproterozoic to lower Cambrian rocks of Australia[a]

Pole	Rock unit	Rock age (Ma)	Lat. (°N)	Long. (°E)	dp (deg)	dm (deg)	*N*	Comment	Reference
Stratified rocks									
19	Hawker Gp base	early Cambrian	−26.7	002.3	8.1	14.3	10	–	Klootwijk (1980)
18	Todd R. Dolomite, Allua & Eninta Fm.	early Cambrian	−43.2	339.9	4.5	7.7	34	u/c test, predates Permo-Carb. folding	Kirschvink (1978)
17	Antrim Plateau Volcanics	early Cambrian	−9	340	17	17	14	reversal in four sites	McElhinny and Luck (1970)
— Neoproterozoic–Cambrian boundary ~540 Ma —									
16	u. Arumbera Sandstone	underlies Cambrian unconformity	−46.6	337.3	4.5	7.7	57	u/c test, predates Permo-Carb. folding	Kirschvink (1978)
15	l. Arumbera SS & u. Pertatataka Fm.	–	−44.3	341.9	7.7	13.6	12	u/c test, predates Permo-Carb. folding	Kirschvink (1978)
14	Pertatataka Fm.–Ringwood Member	–	−65	278	8	15	8	fold test not significant	McWilliams and McElhinny (1980)
13	Bunyeroo Fm.	588±35 Rb–Sr WR shale	−18.1	016.3	6.5	11.8	6	predates Cambro-Ordovician folding	Webb et al. (1983), Schmidt and Williams (1996)
12	Brachina Fm.	–	−45	334	4	7	?	–	Karner and Chamalaun (1979) in McWilliams and McElhinny (1980)
11	Brachina Fm.	–	−33	328	12	20	8	predates Cambro-Ordovician folding	McWilliams and McElhinny (1980)
10	Elatina Fm.	–	−52.4	347.1	3.7	7.4	6	fold test	Schmidt et al. (1991), Schmidt and Williams (1995)
9	Angepena Fm.	724±40 Rb-Sr WR shale	−33	344	9	16	5	cannot demonstrate primary	McWilliams and McElhinny (1980); Webb and Coats (1980) in Preiss (1987)
8	Tapley Hill Fm.	750±53 Rb–Sr WR shale	−39	028	8	15	5	fold test inconclusive — younger overprint?	McWilliams and McElhinny (1980); Webb and Coats (1980) in Preiss (1987)
7	Merinjina Tillite	–	−45	346	6	11	16	cannot demonstrate primary	McWilliams and McElhinny (1980)
6	Copley Qtzite	–	−36	010	7	15	6	younger overprint?	McWilliams and McElhinny (1980)
5	Wooltana Volcanics	827±6 SHRIMP U–Pb on GDS	62	322	16	18	10	Palaeozoic hydro-thermal overprint	McWilliams and McElhinny (1980), M. Idnurm, personal communication, 1997
Mafic intrusive rocks									
MDS*	MDS & Northampton dykes	755±3 SHRIMP U–Pb zircon	−45.8	315.3	4.3	4.3	13	contact test; combined pole	Embleton and Schmidt (1985), this study
YB	YB dykes	probably 750 to 500	−19.9	282.0	28.1	28.1	4	contact test	Giddings (1976)
4*	Kulgera dykes	1054±14 Rb–Sr 1090±32 Sm–Nd	−17	266	12	12	6	predates Permo-Carb. folding	Camacho et al. (1991), Zhao and McCulloch (1993)
3*	Stuart dykes	1076±33 Sm–Nd	+10	262	10	10	?	preliminary pole	Idnurm and Giddings (1988), Zhao and McCulloch (1993)
2*	Lakeview dolerite & IA dykes	1116±12 Rb–Sr	+9.5	311.1	17.4	17.4	10	revised pole	Duff and Embleton (1976), Page (1983), Tanaka and Idnurm (1994)
1*	Giles Complex	1078±5 SHRIMP U–Pb zircon	+18	307	23	29	39	revised pole	Facer (1971), Tanaka and Idnurm (1994), Glikson et al. (1996)

[a] South Pole positions in present-day Australian coordinates. Pole numbers identify data in Figs. 9 and 10. u/c test, unconformity test; WR, whole rock; GDS, Gairdner dyke swarm. Age estimate for Neoproterozoic–Cambrian boundary from Compston et al. (1992), Bowring et al. (1993), Isachsen et al. (1994). Other notes as in Table 5.

6.2. Implications for the Rodinia hypothesis

The Rodinia reconstruction of Powell et al. (1993) preserved the continuity of Grenville-age orogenic belts and optimised the fit between selected palaeopoles from East Gondwana and Laurentia (Fig. 10a). They combined results from the ~730 Ma Malani Rhyolites of India with those

from the YB dykes of Western Australia (MR and YB in Fig. 10) and interpreted the proximity of the mean pole to those for the 723 Ma Franklin dykes and Natkusiak Basalts of northern Canada (L and M) as indicating that East Gondwanaland and Laurentia were still joined at 723 Ma. However, as discussed elsewhere (Veevers et al., 1997; Wingate et al., 1998), this assessment is not necessarily rigorous, owing mainly to poor age control for the Australian and Indian poles and imprecision of the YB pole ($A_{95}=28°$). If Australia and Laurentia were joined until at least 723 Ma, as suggested by Powell et al. (1993), then the 755 Ma palaeopoles for each continent should coincide. However, the MDS pole for Australia at 755 Ma lies at least 30° away from the 780 to 740 Ma APW path segment for Laurentia [Fig. 10(a)]. There are two possible explanations for this discrepancy: either Australia and Laurentia were in relative motion prior to 755 Ma, or the early Neoproterozoic configuration of the two continents was different from the reconstruction of Powell et al. (1993).

The latest Mesoproterozoic to Early Cambrian palaeolatitude curves and APW paths for Australia and Laurentia shown in Figs. 10 and 11 are calibrated with selected 'key' data (Table 5). For Laurentia, key data (large triangles) fulfil the following criteria: (1) precise and reliable ages are available; (2) the magnetisation is inferred, from field or laboratory stability tests, to be primary; (3) a sufficient number of samples has been analysed; (4) structural correction is unambiguous. The key poles are superimposed on the Laurentian APW path as interpreted by Irving and McGlynn (1981), Park and Jefferson (1991), Hyodo and Dunlop (1993), Park (1994), and others; note that the APW path for Laurentia is based on many more data than are shown here. Five poles shown by small triangles represent data for which age control is less reliable (details in Table 5). The large error (±40 Ma) assigned to the age of pole K (based on overprints in the Tsezotene and Thundercloud Formations) reflects uncertainty as to whether or not it predates or postdates the 723 Ma Franklin poles (L and M), as discussed by Park and Jefferson (1991). It is assumed here that the overprint pole K predates the 723 Ma

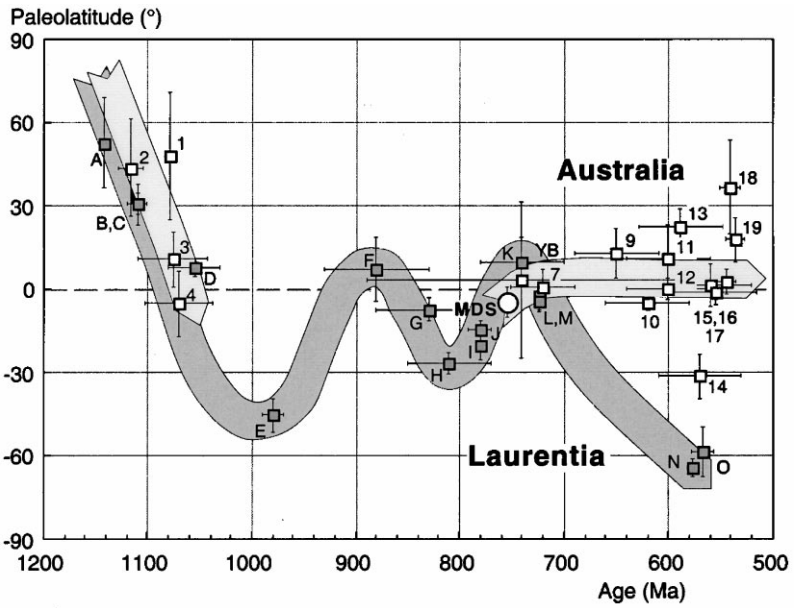


Fig. 11. Variation of palaeolatitude with time, based on palaeomagnetic data from Australia and Laurentia (Tables 5 and 6). Values are calculated for reference locality Winnipeg, after rotating Laurentia into a Rodinia fit with Australia. Note the absence of data for Australia between ~1050 and 755 Ma. Palaeolatitudes and ages are shown with 95% confidence intervals.

Franklin poles (L and M). This choice minimises the angular difference between the MDS pole and the Laurentian APW path and makes our assessment more conservative; adopting the alternative interpretation would increase the discrepancy by an additional 10 to 15°.

There are few palaeomagnetic results from Australian rocks for the same period of time (Table 6), and no data for the interval ~1070 to 755 Ma. The selection criteria for Australian data are therefore less stringent, and the four key poles (in addition to MDS) employed for Australia (large circles in figures) are simply those for which the rocks have been dated radiometrically (excluding Rb–Sr shale results); all remaining data are shown with small symbols. Poles 1, 2, and 4 were interpreted by the original authors to reflect primary magnetisations and to require no structural correction, although none has been tested rigorously.

The angular difference of $\geq 30°$ between the MDS pole and the 780 to 740 Ma Laurentia APW path segment in the reconstruction of Powell et al. (1993) is substantial, and is considered to be highly significant. Although it is conceivable that the APW path for Laurentia made an unrecognised excursion towards the MDS pole at 755 Ma, there are no data to suggest this might be the case. Palaeolatitudes for Australia and Laurentia at 755 Ma remain consistent with a Rodinia configuration (Fig. 11); the discrepancy in pole positions results mainly from a 30° counterclockwise rotation of Australia with respect to Laurentia [Fig. 10(b)]. If the Powell et al. (1993) reconstruction is correct, these observations are evidence for relative motion between Australia and Laurentia prior to 755 Ma.

In a correct reconstruction of Australia and Laurentia at 755 Ma, the palaeopoles should overlap. To evaluate possible configurations that satisfy the available palaeomagnetic data, the APW paths and continents are rotated so that the MDS pole, and an interpolated pole position for Laurentia at 755 Ma (estimated assuming a constant rate of APW between 780 and 740 Ma), are coincident at the south pole of the projection [Fig. 10(b)]. In this palaeomagnetically constrained fit, the Grenville Province of Laurentia intersects a region of Antarctica, the Transantarctic Mountains, from which there are no reported occurrences of Grenville-age rocks. Secondly, the APW paths do not match for earlier periods of time during which the continents are thought to have been joined. In the reconstruction of Powell et al. (1993), the ~1070 Ma Kulgera dykes pole (4) for Australia is 12° away from, and within 95% confidence limits of, the 1055 Ma pole (D) for Laurentia, but when the 755 Ma poles are superimposed this difference increases significantly to 25°. The 1100 Ma poles for the two continents move apart by a similar amount. Finally, the Adelaidean sedimentary sequences of Australia are placed at least 2500 km away from potentially equivalent strata in western Laurentia, making their correlation less likely, although ample space is left to incorporate additional continental fragments, such as South China (Li et al., 1996), in the gap between Australia and Laurentia.

Correlation between the Rapitan tillites of the Mackenzie Mountains in western Canada and the Sturtian tillites of Australia has been advocated by Eisbacher (1985), Bell and Jefferson (1987), Young (1992), and others. A U–Pb zircon date of 755 ± 18 Ma for a granitic clast (Ross and Villeneuve, 1997) within the Rapitan tillite provides a firm older limit for the Rapitan glaciation. It was suggested by Powell et al. (1994) that the Sturtian glaciation in Australia preceded or was broadly synchronous with breakup — if this is correct, then a corollary of the present study is that the Sturtian rocks should be older than 755 Ma and hence may not be coeval with the Rapitan Group of Laurentia.

The results of this study indicate that a Rodinian connection between Australia and Laurentia may not have existed at 755 Ma. If the reconstruction of Powell et al. (1993) is correct, the data imply that Australia and Laurentia were in relative motion prior to 755 Ma. This would require continental breakup to have occurred before 755 Ma, although the similarity in palaeolatitudes indicates that the continents could have remained in close proximity to each other. This proposed younger limit for breakup is ≥30 million years older than the 700 to 720 Ma age proposed by Powell et al. (1993, 1994), and ~200 million

years older than the 560 Ma age of breakup suggested by Veevers et al. (1997). Further tests of the Australia–Laurentia connection must await additional well-dated palaeomagnetic results for Australia and Laurentia between ~1050 and 723 Ma.

Acknowledgements

This work was undertaken partly at the Research School of Earth Sciences, The Australian National University, supported by an Overseas Postgraduate Research Scholarship and an ANU Ph.D. Scholarship to MW, and partly at the Tectonics Special Research Centre, Department of Geology and Geophysics, The University of Western Australia, supported by an Australian Research Council Postdoctoral Fellowship and small grant to MW. Fieldwork conducted during this project was supported in part by the Geological Survey of Western Australia. Palaeomagnetic analyses were carried out at the Black Mountain laboratory, Canberra, a joint facility of the Australian Geological Survey Organisation and the Australian National University. J.N. Connelly, R. Ernst, S.S. Harlan, M. Idnurm, D.L. Jones, Z.-X. Li, C.McA. Powell, I.M. Tyler, and J.J. Veevers are thanked for their valuable critiques of the manuscript. JWG publishes with authority of the Executive Director, Australian Geological Survey Organisation. Tectonics Special Research Centre Publication No. 53.

References

Aleinikoff, J.N., Zartman, R.E., Walter, M., Rankin, D.W., Lyttle, P.T., Burton, W.C., 1995. U–Pb ages of metarhyolites of the Catoctin and Mount Rogers formations, Central and Southern Appalachians, evidence for two pulses of Iapetan rifting. Am. J. Sci. 295, 428–454.

Bell, R.T., Jefferson, C.W., 1987. An hypothesis for an Australian–Canadian connection in the Late Proterozoic and the birth of the Pacific Ocean. Proc. Pacific Rim Congress '87, Parkville, Australia.

Bickle, M.J., Bettenay, L.F., Barley, M.E., Chapman, H.J., Groves, D.I., Campbell, I.H., de Laeter, J.R., 1983. A 3500 Ma plutonic and volcanic calc-alkaline province in the Archean East Pilbara Block. Contrib. Mineral. Petrol. 84, 25–35.

Bickle, M.J., Bettenay, L.F., Chapman, H.J., Groves, D.I., McNaughton, N.J., Campbell, I.H., de Laeter, J.R., 1989. The age and origin of younger granitic plutons of the Shaw Batholith in the Archaean Pilbara Block, Western Australia. Contrib. Mineral. Petrol. 101, 361–376.

Blake, T.S., McNaughton, N.J., 1984. A geochronological framework for the Pilbara region in: University of Western Australia Geology Department and University Extension Publ. 9, 1–22.

Borg, S.G., DePaolo, D.J., 1994. Laurentia, Australia, and Antarctica as a Late Proterozoic supercontinent: constraints from isotopic mapping. Geology 22, 307–310.

Bowring, S.A., Grotzinger, J.P., Isachsen, C.E., Knoll, A.H., Pelechaty, S.M., Kolosov, P., 1993. Calibrating rates of early Cambrian evolution. Science 261, 1293–1298.

Boyd, D.M., Tucker, D.H., 1990. Australian magnetic dykes. In: Parker, A.J., Rickwood, P.C., Tucker, D.H. (Eds.), Mafic Dykes and Emplacement Mechanisms. Balkema, Rotterdam, pp. 391–399.

Brookfield, M.E., 1993. Neoproterozoic Laurentia–Australia fit. Geology 21, 683–686.

Buchan, K.L., 1978. Magnetic overprinting in the Thanet gabbro complex, Ontario. Can. J. Earth Sci. 15, 1407–1421.

Buchan, K.L., Dunlop, D.J., 1976. Paleomagnetism of the Haliburton Intrusions: superimposed magnetisations, metamorphism, and tectonics in the late Precambrian. J. Geophys. Res. 81, 2951–2967.

Camacho, A., Simons, B., Schmidt, P.W., 1991. Geological and paleomagnetic significance of the Kulgera Dyke Swarm, N.T., Australia. Geophys. J. Int. 107, 37–45.

Claoué-Long, J.C., Compston, W., Roberts, J., Fanning, C.M., 1995. Two Carboniferous ages: a comparison of SHRIMP zircon ages with conventional zircon ages and $^{40}Ar/^{39}Ar$ analysis. In: Berggren, W.A., Kent, D.V., Aubrey, M.-P., Hardenbol, J. (Eds.), Geochronology, Time Scales, and Global Stratigraphic Correlation. S.E.P.M. Special Publication 54, 3–21.

Compston, W., Williams, I.S., Kirschvink, J.L., Zhang, Z., Max, G., 1992. Zircon U–Pb ages for the Early Cambrian time-scale. J. Geol. Soc. London 149, 171–184.

Condon, M.A., 1955. Minilya — 4 Mile Geological Series Sheet F50-13 & F49-16 and Explanatory Notes. Bur. Min. Res., Canberra. 19 pp.

Condon, M.A., 1962. Kennedy Range — 1:250,000 Geological Series Sheet SG50-1 and Explanatory Notes. Bur. Min. Res., Canberra. 28 pp.

Dalziel, I.W.D., 1991. Pacific margins of Laurentia and East Antarctica–Australia as a conjugate rift pair: evidence and implications for an Eocambrian supercontinent. Geology 19, 598–601.

Davis, D.W., Sutcliffe, R.H., 1985. U–Pb ages from the Nipigon plate and Northern Lake Superior. Geol. Soc. Am. Bull. 96, 1572–1579.

Duff, B.A., Embleton, B.J.J., 1976. Paleomagnetic directions in

Precambrian basic intrusives of the Mt. Isa Province, Australia. Earth Planet. Sci. Lett. 28, 418–426.
Eisbacher, G.H., 1985. Late Proterozoic rifting, glacial sedimentation and sedimentary cycles in the light of Windermere deposition, western Canada. Palaeogeogr. Palaeoclim. Palaeoecol. 51, 231–254.
Embleton, B.J.J., Schmidt, P.W., 1985. Age and significance of magnetisations in dolerite dykes from the Northampton Block, Western Australia. Aust. J. Earth Sci. 32, 279–286.
Ernst, R.E., Buchan, K.L, 1993. Paleomagnetism of the Abitibi dyke swarm, southern Superior Province, and implications for the Logan Loop. Can. J. Earth Sci. 30, 1886–1897.
Facer, R.A., 1971. Magnetic properties of rocks from the Giles Complex, central Australia. J. Proc. R. Soc. N.S.W. 104, 45–61.
Giddings, J.W., 1976. Precambrian paleomagnetism in Australia I: basic dykes and volcanics from the Yilgarn Block. Tectonophysics 30, 91–108.
Giddings, J.W., Klootwijk, C.T., Rees, J., Groenewoud, A., 1997. Automated AF-demagnetisation on the 2-G Enterprises through-bore, cryogenic magnetometer. Geol. Mijnbouw 76, 35–44.
Glikson, A.Y., Stewart, A.J., Ballhaus, C.G., Clarke, G.L., Feeken, E.H.J., Leven, J.H., Sheraton, J.W., Sun, S.-S., 1996. Geology of the western Musgrave Block, central Australia, with particular reference to the mafic–ultramafic Giles Complex. Aust. Geo. Surv. Org. Bull. 239. 206 pp.
Gose, W.A., Helper, M.A., Connelly, J.N., Hutson, F.E., Dalziel, I.W.D., 1997. Paleomagnetic data and U–Pb isotopic age determinations from Coats Land, Antarctica: implications for late Proterozoic reconstructions. J. Geophys. Res. 102, 7887–7902.
Graham, K.W.T., 1961. The magnetisation of a surface outcrop by lightning currents. J. R. Astron. Soc. 6, 85–102.
Halls, H.C., Pesonen, L.J., 1982. Paleomagnetism of Keweenawan rocks. In: Geol. Soc. Am. Mem. 156, 173–201.
Harlan, S.S., Geissman, J.W., Snee, L.W., 1997. Paleomagnetic and 40Ar/39Ar geochronologic data from late Proterozoic mafic dykes and sills, Montana and Wyoming. U.S.G.S. Prof. Paper 1580. 16 pp.
Heaman, L.M., LeCheminant, A.N., Rainbird, R.H., 1991. 0.78 Ga U–Pb age for mafic magmatism in the Mackenzie Mountains: precise distinction from 0.72 Ga Franklin events. Cordilleran Geology and Exploration Roundup, Program and Abstracts, Vancouver.
Heaman, L.M., LeCheminant, A.N., Rainbird, R.H., 1992. Nature and timing of Franklin igneous events, Canada: implications for a Late Proterozoic mantle plume and the break-up of Laurentia. Earth Planet. Sci. Lett. 109, 117–131.
Henry, S.G., Mauk, F.J., Van der Voo, R., 1977. Paleomagnetism of the upper Keweenawan sediments: the Nonesuch Shale and Freda Sandstone. Can. J. Earth Sci. 14, 1128–1138.
Hickman, A.H., Lipple, S.L., 1978. Marble Bar, Western Australia. In: Western Australia Geological Survey, 1:250,000 Geological Series Explanatory Notes, 24 pp.
Hinthorne, J.R., Andersen, C.A., Conrad, R.L., Lovering, J.F., 1979. Single-grain $^{207}Pb^{*}/^{206}Pb^{*}$ and U/Pb age determinations with a 10 mm spatial resolution using the ion microprobe mass analyzer (IMMA). Chem. Geol. 25, 271–303.
Hocking, R.M., 1990. Carnarvon Basin. In: Geology and Mineral Resources of Western Australia. Geological Survey of Western Australia Memoir 3, 457–495.
Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwanaland inside-out? Science 252, 1409–1412.
Hyodo, H., Dunlop, D.J., 1993. Effect of anisotropy on the paleomagnetic contact test for a Grenville dyke. J. Geophys. Res. 98, 7997–8017.
Idnurm, M., Giddings, J.W., 1988. Australian Precambrian polar wander: a review. Precamb. Res. 40, 61–88.
Irving, E., McGlynn, J.C., 1981. On the coherence rotation and paleolatitude of Laurentia in the Proterozoic. In: Kroner, A. (Ed.), Precambrian Plate Tectonics. Elsevier, pp. 561–598.
Isachsen, C.E., Bowring, S.A., Landing, E., Samson, S.D., 1994. New constraint on the division of Cambrian time. Geology 22, 496–498.
Jefferson, C.W., 1978. Correlation of Middle and Upper Proterozoic strata between northwestern Canada and south and central Australia. In: Geological Society of America Prog. Abstr. 7, 429.
Kirschvink, J.L., 1978. The Precambrian–Cambrian boundary problem: paleomagnetic directions from the Amadeus Basin, central Australia. Earth Planet. Sci. Lett. 40, 91–100.
Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.
Klootwijk, C.T., 1980. Early Paleozoic paleomagnetism in Australia. Tectonophysics 64, 249–332.
Krogh, T.E., Corfu, F., Davis, D.W., Dunning, G.R., Heaman, L.M., Kamo, S.L., Machado, N., Greenough, J.D., Nakamura, E., 1987. Precise U–Pb isotopic ages of diabase dykes and mafic to ultramafic rocks using trace amounts of baddeleyite and zircon. In: Geological Association of Canada Spec. Paper 34, 147–152.
Li, Z.X., Zhang, L., Powell, C.M., 1996. Positions of the East Asian cratons in the Neoproterozoic supercontinent Rodinia. Aust. J. Earth Sci. 43, 593–604.
Libby, W.G., de Laeter, J.R., Myers, J.S., 1986. Geochronology of the Gascoyne Province. Western Australia Geological Survey Report 20. 31 pp.
Mattinson, J.M., Graubard, C.M., Parkinson, D.L., McClelland, W.C., 1996. U–Pb reverse discordance in zircons: the role of fine-scale oscillatory zoning and sub-micron transport of Pb. In: Basu, A., Hart, S. (Eds.), Earth Processes: Reading the Isotopic Code. American Geophysical Union Geophys. Monograph 95, 355–370.
McElhinny, M.W., Luck, G.R., 1970. The paleomagnetism of the Antrim Plateau Volcanics of Northern Australia. Geophys. J. R. Astron. Soc. 20, 191–205.
McLaren, A.C., Fitz Gerald, J.D., Williams, I.S., 1994. The microstructure of zircon and its influence on the age determination from Pb/U isotopic ratios measured by ion microprobe. Geochim. Cosmochim. Acta 58, 993–1005.

McWilliams, M.O., McElhinny, M.W., 1980. Late Precambrian paleomagnetism of Australia, the Adelaide Geosyncline. J. Geol. 88, 1–26.

Meert, J.G., Van der Voo, R., Payne, T.W., 1994. Paleomagnetism of the Catoctin volcanic province: a new Vendian–Cambrian apparent polar wander path for North America. J. Geophys. Res. 99, 4625–4641.

Moores, E.M., 1991. Southwest U.S.–East Antarctic (SWEAT) connection: a hypothesis. Geology 19, 425–428.

Muhling, P.C., Brakel, A.T., 1985. Geology of the Bangemall Group — the evolution of an intracratonic Proterozoic basin. In: Geological Survey Western Australia Bulletin 128, 1–219.

Nelson, D.R., 1995. Compilation of SHRIMP U–Pb zircon data, 1994. Geological Survey of Western Australia Record 1995/3. 244 pp.

Page, R.W., 1983. Chronology of magmatism, skarn formation and uranium mineralisation, Mary Kathleen, Queensland, Australia. Econ. Geol. 78, 838–853.

Palmer, H.C., Baragar, W.R.A., Fortier, M., Foster, J.H., 1983. Paleomagnetism of Late Proterozoic rocks, Victoria Island, Northwest Territories, Canada. Can. J. Earth Sci. 20, 1456–1469.

Park, J.K., 1981a. Analysis of the multicomponent magnetisation of the Little Dal Group, Mackenzie Mountains, Northwest Territories, Canada. J. Geophys. Res. 86, 5134–5146.

Park, J.K., 1981b. Paleomagnetism of the Late Proterozoic sills in the Tsezotene Formation, Mackenzie Mountains, Northwest Territories, Canada. Can. J. Earth Sci. 18, 1572–1580.

Park, J.K., 1989. Paleomagnetism and the origin of the Mackenzie Arc of northwestern Canada. Can. J. Earth Sci. 26, 2194–2203.

Park, J.K., 1994. Paleomagnetic constraints on the position of Laurentia from middle Neoproterozoic to early Cambrian times. Precamb. Res. 69, 95–112.

Park, J.K., Aitken, J.D., 1986a. Paleomagnetism of the Late Proterozoic Tsezotene Formation of northwestern Canada. J. Geophys. Res. 91, 4955–4970.

Park, J.K., Aitken, J.D., 1986b. Paleomagnetism of the Katherine Group in the Mackenzie Mountains: implications for post-Grenville (Hadrynian) apparent polar wander. Can. J. Earth Sci. 23, 308–323.

Park, J.K., Jefferson, C.W., 1991. Magnetic and tectonic history of the Late Proterozoic Upper Little Dal and Coates Lake Groups of northwestern Canada. Precamb. Res. 52, 1–35.

Park, J.K., Buchan, K.L., Harlan, S.S., 1995. A proposed giant radiating dyke swarm fragmented by the separation of Laurentia and Australia based on paleomagnetism of ca. 780 Ma mafic intrusions in western North America. Earth Planet. Sci. Lett. 132, 129–139.

Parker, A.J., Rickwood, P.C., Baillie, P.W., Boyd, D.M., McClenaghan, M.P., Freeman, M.J., Pietsch, B.A., Murray, C.G., Myers, J.S., 1987. Mafic dyke swarms of Australia. In: Halls, H.C., Fahrig, W.F. (Eds.), Mafic Dyke Swarms, Geological Association of Canada Spec. Paper 34, 401–418.

Pesonen, L.J., 1979. Paleomagnetism of late Precambrian Keweenawan igneous and baked contact rocks from the Thunder Bay District, northern Lake Superior. Geol. Soc. Finland Bull. 51, 27–44.

Powell, C.McA., Roots, S.R., Veevers, J.J., 1988. Pre-breakup continental extension in East Gondwanaland and the early opening of the eastern Indian Ocean. Tectonophysics 155, 261–283.

Powell, C.McA., Li, Z.X., McElhinny, M.W., Meert, J.G., Park, J.K., 1993. Paleomagnetic constraints on timing of the Neoproterozoic breakup of Rodinia and the Cambrian formation of Gondwana. Geology 21, 889–892.

Powell, C.McA., Preiss, W.V., Gatehouse, C.G., Krapez, B., Li, Z.X., 1994. South Australian record of a Rodinian epicontinental basin and its mid-Neoproterozoic breakup ($\sim$700 Ma) to form the palaeo-Pacific Ocean. Tectonophysics 237, 113–140.

Preiss, W.V., 1987. The Adelaide Geosyncline — Late Proterozoic stratigraphy sedimentation paleontology and tectonics. Geological Survey of South Australia Bulletin 53. 438 pp.

Pullaiah, G., Irving, E., Buchan, K.L., Dunlop, D.J., 1975. Magnetisation changes caused by burial and uplift. Earth Planet. Sci. Lett. 28, 133–143.

Ross, G.M., Villeneuve, M.E., 1997. U–Pb geochronology of strange stones in Neoproterozoic diamictites Canadian Cordillera: implications for provenance and ages of deposition. In: Radiogenic Age and Isotope Studies, Report 10. Geological Survey of Canada Curr. Res. 1997-F, 141–155.

Ross, G.M., Parrish, R.R., Winston, D., 1992. Provenance and U–Pb geochronology of the Middle Proterozoic Belt Supergroup (northwestern United States): implications for the age of deposition and pre-Panthalassa plate reconstructions. Earth Planet. Sci. Lett. 113, 57–76.

Schmidt, P.W., 1993. Paleomagnetic cleaning strategies. Phys. Earth Planet. Int. 76, 169–178.

Schmidt, P.W., Williams, G.E., 1995. The Neoproterozoic climatic paradox: equatorial paleolatitude for Marinoan glaciation near sea level in South Australia. Earth Planet. Sci. Lett. 134, 107–124.

Schmidt, P.W., Williams, G.E., 1996. Paleomagnetism of the ejecta-bearing Bunyeroo Formation, late Neoproterozoic, Adelaide fold belt, and the age of the Acraman impact. Earth Planet. Sci. Lett. 144, 347–357.

Schmidt, P.W., Williams, G.E., Embleton, B.J.J., 1991. Low paleolatitude of Late Proterozoic glaciation: early timing of remanence of hematite of the Elatina Formation, South Australia. Earth Planet. Sci. Lett. 105, 355–367.

Schwarz, E.J., 1977. Depth of burial from remanent magnetisation: the Sudbury irruptive at the time of diabase intrusion (1250 Ma). Can. J. Earth Sci. 14, 82–88.

Schwarz, E.J., Buchan, K.L., 1989. Identifying types of remanent magnetisation in igneous contact zones. Phys. Earth Planet. Int. 58, 155–162.

Symons, D.T.A., Chiasson, A.D., 1991. Paleomagnetism of the Callander Complex and the Cambrian apparent polar wander path for North America. Can. J. Earth Sci. 28, 355–363.

Tanaka, H., Idnurm, M., 1994. Paleomagnetism of Proterozoic

mafic intrusions and host rocks of the Mt. Isa Inlier, Australia. Precamb. Res. 69, 241–258.

Tyler, I., 1990. Mafic dyke swarms. In: Geology and Mineral Resources of Western Australia. Geological Survey of Western Australia Memoir 3, 191–195.

Veevers, J.J., Walter, M.R., Scheibner, E., 1997. Neoproterozoic tectonics of Australia–Antarctica and Laurentia and the 560 Ma birth of the Pacific Ocean reflect the 400 m.y. Pangean Supercycle. J. Geol. 107, 225–242.

Walton, D., 1980. Time–temperature relations in the magnetisation of assemblies of single-domain grains. Nature 286, 245–247.

Webb, A.W., Coats, R.P., Fanning, C.M., Flint, R.B., 1983. Geochronological framework of the Adelaide Geosyncline. In: Geological Society of Australia Abstract 10, 7–9.

Williams, I.R., 1990. Bangemall Basin. In: Geology and Mineral Resources of Western Australia. Western Australia Geological Survey Memoir 3, 308–329.

Wingate, M.T.D., 1997. Testing Precambrian continental reconstructions using ion microprobe U–Pb baddeleyite geochronology and paleomagnetism of mafic igneous rocks. Ph.D. Thesis, Australian National University, 236 pp.

Wingate, M.T.D., Campbell, I.H., Compston, W., Gibson, G.M., 1998. Ion microprobe U–Pb ages for Neoproterozoic basaltic magmatism in south-central Australia and implications for the breakup of Rodinia. Precamb. Res. 87, 135–159.

Young, G.M., 1992. Late Proterozoic stratigraphy and the Canada–Australia connection. Geology 20, 215–218.

Zhao, J.-X., McCulloch, M.T., 1993. Sm–Nd mineral isochron ages of Late Proterozoic dyke swarms in Australia: evidence for two distinct events of mafic magmatism and crustal extension. Chem. Geol. 109, 341–354.